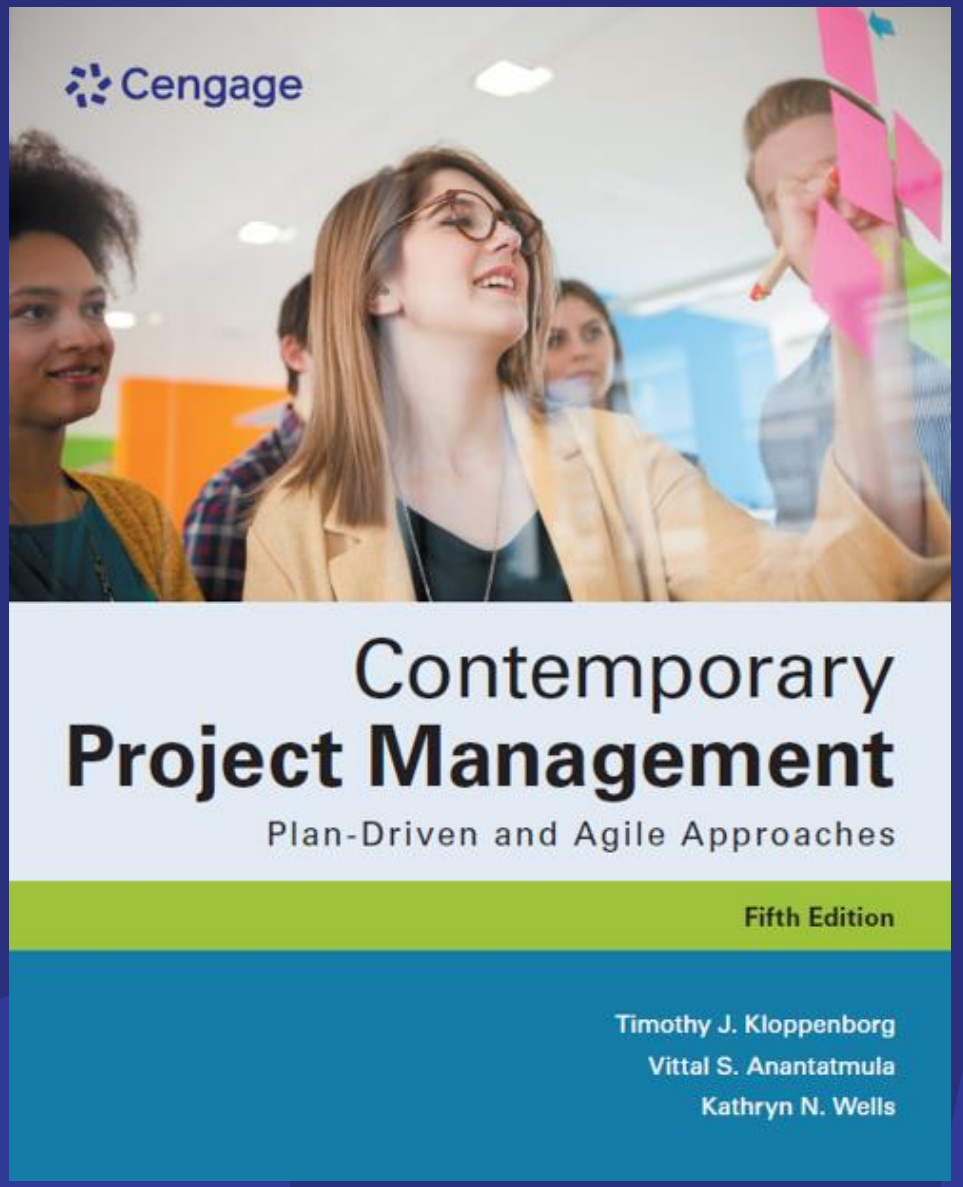



# Chapter 9

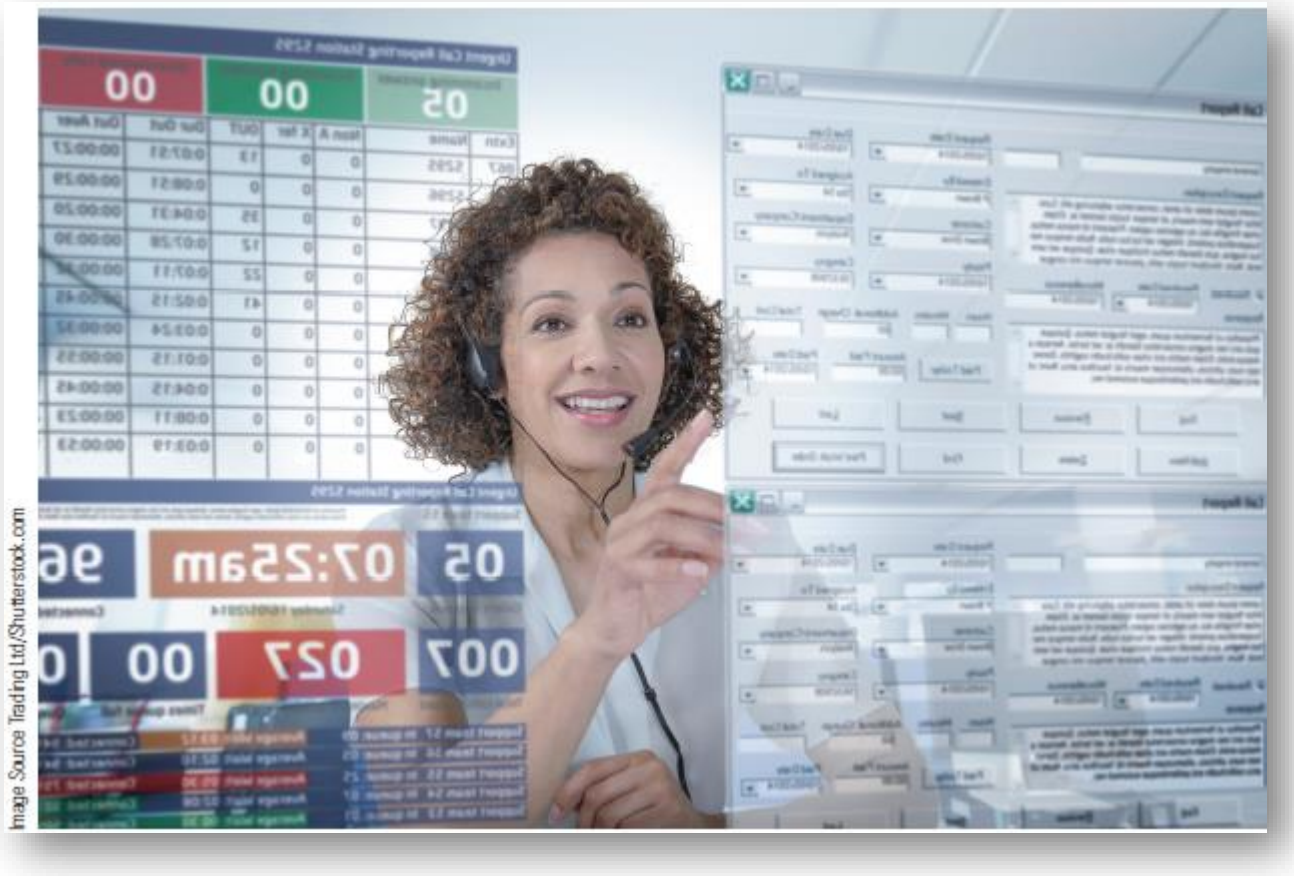
## Resourcing and Accelerating Projects

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# Icebreaker

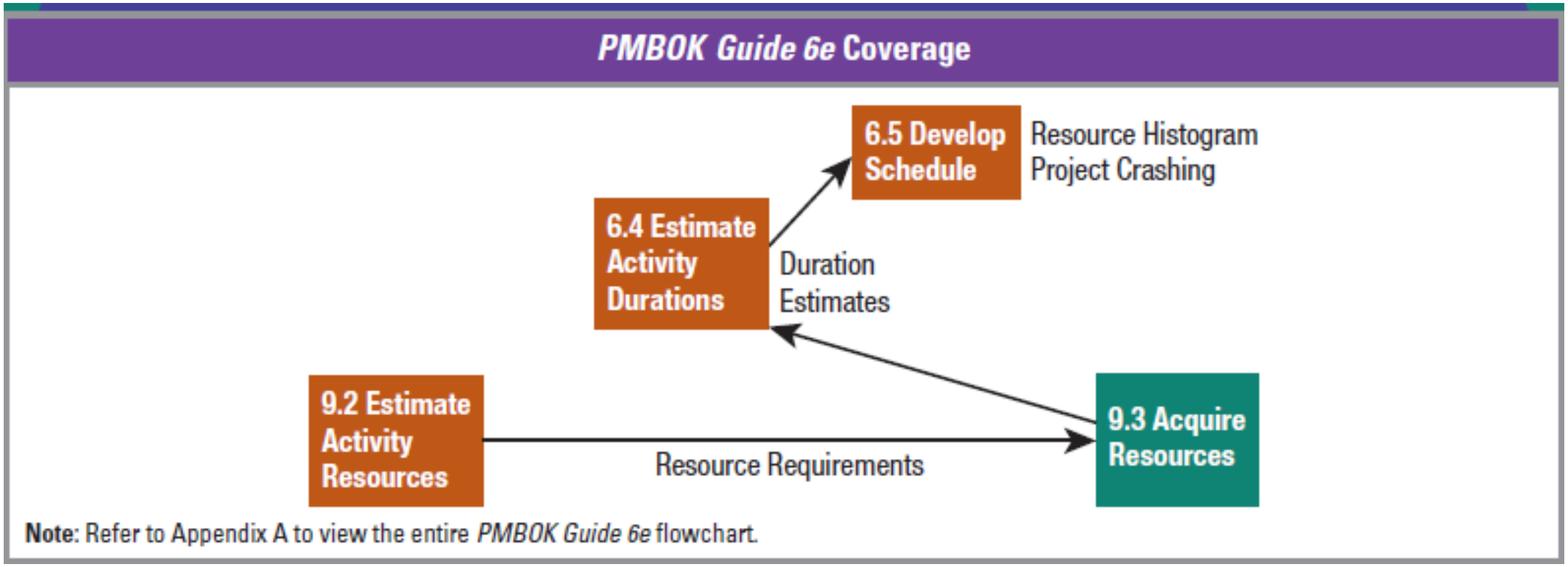
What are two things you learned from Scheduling Projects?





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# Resourcing and Accelerating Projects (1 of 2)





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# Resourcing and Accelerating Projects (2 of 2)

| Overview of the Book |
| --- |
| 9.1 Abilities Needed When Resourcing Projects |
| 9.2 Estimate Resource Needs |
| 9.3 Plan Resource Management |
| 9.4 Project Team Composition Issues |
| 9.5 Assign a Resource to Each Activity |
| 9.6 Dealing with Resource Overloads |
| 9.7 Compress the Project Schedule |
| 9.8 Alternative Scheduling Methods |
| 9.9 *PMBOK Guide 7e* |
| 9.10 Agile Project Resourcing |
| 9.11 Using MS Project for Resource Allocation |
| 9.12 Summary |


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# Core Objectives

- **9-1** Assign resources on a RACI chart, Gantt chart, and resource histogram.
- **9-2** Develop an effective project schedule with resources and resource constraints.
- **9-3** Compress a project schedule using crashing and fast tracking.
- **9-4** Compare various scheduling methods.
- **9-5** Assign resources, pinpoint overloads, and describe methods of dealing with them using MS Project.
- **9-6** Create a human resources management plan and a staffing management plan.


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# Agile Objectives

- **9-7** Describe Agile concepts and tools that enable teams to deliver value quickly.


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# 9.1 Abilities Needed When Resourcing Projects

| |
|---|
| 9.1a The Science and Art of Resourcing Projects |
| 9.1b Considerations When Resourcing Projects |
| 9.1c Activity- versus Resource-Dominated Schedules |


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# 9.1 Behavioral Skills Needed by a Project Manager

- Selecting people who work well in teams.
- Identifying exactly what each person needs to accomplish *(individually and collectively).*
- Ensuring that each person has the desire to improve.
- Dealing with difficult individual work schedules.
- Getting people to work overtime when there are schedule conflicts.
- Making honest and open estimates of the amount of work required.
- Assembling an effective team.
- Dealing with people from diverse backgrounds.
- Deciding where each person will work.
- Deciding how a geographically dispersed team can work virtually and effectively.


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# 9.1a The Science and Art of Resourcing Projects

- A resource-based schedule that is technically brilliant but has little acceptance from those who must do the work has little value.
- A motivated team with poor assignments or very challenging assignments is more likely to be a better option for success.


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# 9.1b Considerations When Resourcing Projects

- Projects always have trade-offs (*with respect to resources, trade-offs among schedule, human resources, costs, quality, and scope).*
- Resources are limited, and Project Managers must recognize resource limitations to prevent overpromising.
- People often constitute a large portion of the total project cost.
- Managing people is the most challenging aspect of a project.
- Resource availability, specifically human resources, is a major constraint for projects.


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# 9.1c Activity- versus Resource-Dominated Schedules

The schedule can be constrained in two ways: one is based more on the activities *(scope-constrained schedule)* or the resource limits *(resource-constrained schedule)*, which may be cost-constrained by budget availability.


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# Exhibit 9.1

## Activity- versus Resource-Dominated Schedule Basis Comparison

| | More on Activity | More on Resource |
|---|---|---|
| **Time in project when scope is determined** | Early | Late |
| **Confidence in duration estimates** | High | Low |
| **Rate of resource learning** | Small | Extensive |
| **Specialization of resources** | Commodity | Unique |
| **Availability of resources** | Easily available | Tight availability |
| **Firmness of activity predecessors (order)** | Absolute | Optional |
| **Concurrency of activities** | Little | Significant |


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# 9.2 Estimate Resource Needs (1 of 2)

| |
|---|
| Estimating Activity Resources |
| |
| |


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# 9.2 Estimate Resource Needs (2 of 2)

**Estimating Activity Resources** is a process of assessing all types of resources—people, materials, tools, and equipment (*along with quantities)*—required for each activity to complete it as specified in project scope.


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# 9.3 Plan Resource Management (1 of 2)

| |
|---|
| 9.3a Identify Potential Resources |
| 9.3b Determine Resource Availability |
| 9.3c Decide Timing Issues When Resourcing Projects |


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# 9.3 Plan Resource Management (2 of 2)

- **Plan Resource Management**
  is the process of identifying resources and required skills for a project, defining and assigning roles and responsibilities to all resources, developing a reporting hierarchy, and communicating expectations.
- **Staffing Management Plan**
  is a proposal, focused on acquiring, developing, and retaining human resources for as long as you need them on the project.


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# Exhibit 9.2

**Resource Planning at Atos**

Adequate resourcing is perhaps the most critical activity required on a project at Atos.

It starts with a successful estimation while bidding for a contract. Levers such as onshore/offshore mix, specific location, skill required on a year-by-year basis (for long-term engagements), current availability of staff, and ease of recruitment are all considerations while staffing projects. Based on these parameters, internal placements/movements are first done. If there are any gaps in skills, recruitment is initiated. The timeline for recruiting is taken into consideration for resource planning as well. For example, in India, hiring from the market could take anywhere from two to four months. In Mexico, it is usually six to eight weeks, whereas in the United States, it could be as low as three weeks. Global mobility of resources within Atos is heavily encouraged, especially for key roles. If it is determined that the skill set required cannot be available on time, for whatever reason, subcontracting is an acceptable alternative. Resource costs are a big part of the decision made to allocate people. For example, if work requires client site presence, it is impractical to assign a resource that lives in California to work for a project in New York. (Travel costs would be too heavy).

Resource Planning is emphasized so that required skills can be forecasted and made available as projects/contracts are won. Proactive investments in niche or growing areas are made as needed. Resource planning is done yearly, with a monthly review to align with changing business needs. There is an emphasis on training and job shadowing while resources are waiting for work. Graduate programs are emphasized in most countries as well, thereby ensuring a good mix of available resources for projects.

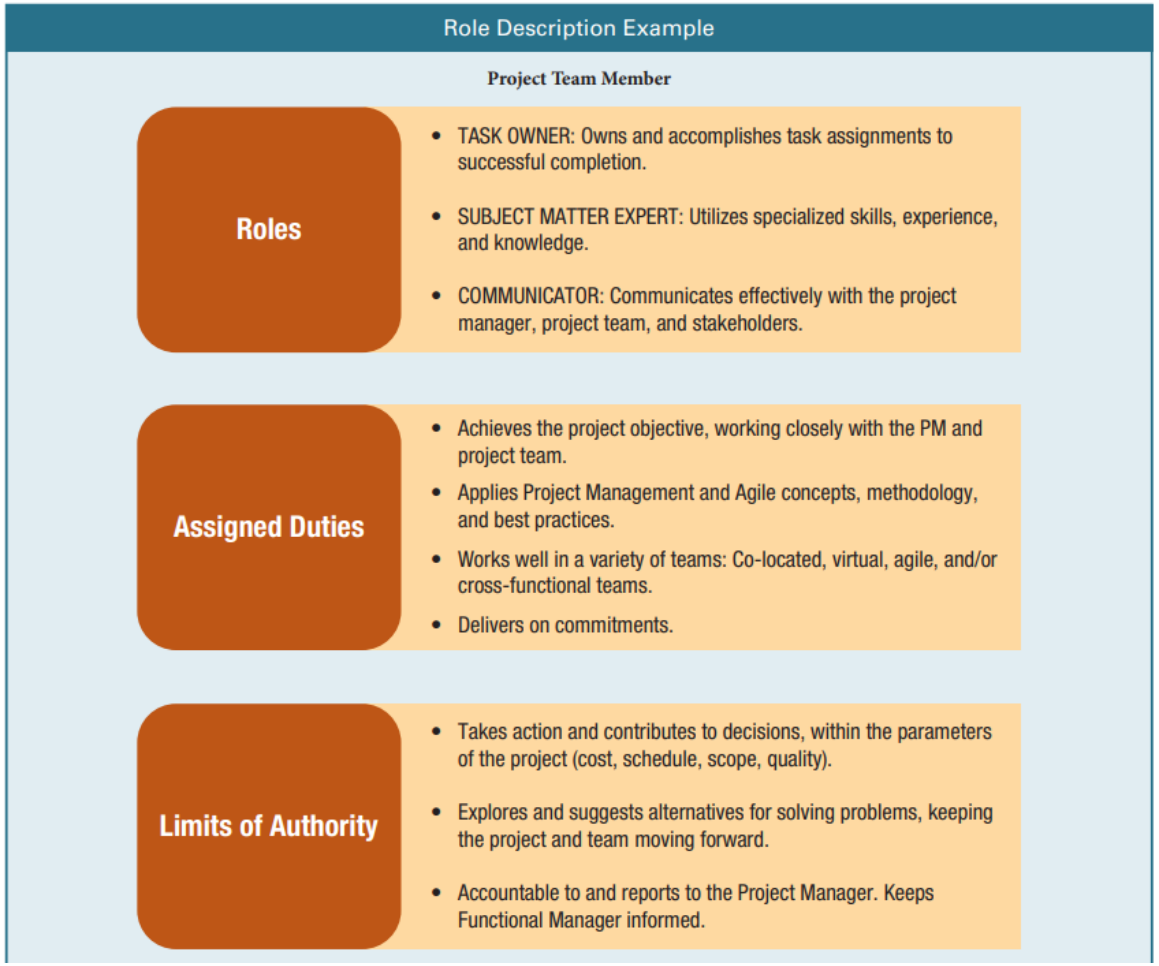
Resourcing strategies are visited quarterly, e.g.: what type of skills should we invest in versus partner on; which locations should be our focus; what level of experience do we need within the organization; how can we pay competitively to attract the best talent; and how do we keep employees happy so that they can continue to grow.

The ultimate aim is to ensure that Atos can deliver on the commitments made to the client for the project timeline and deliverables, and this is not possible without the right resources at the right time, place, quality, and cost. Client satisfaction is one of the most important KPIs (key performance indicators) of any consulting company, and it often is directly a result of having the right people delivering what was promised.

*Source:* Rachana Sampat (*MBA, PMP)*, Head of Workforce Management, Atos, North America.


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# Exhibit 9.3



Role Description Example

**Project Team Member**

**Roles**

- TASK OWNER: Owns and accomplishes task assignments to successful completion.
- SUBJECT MATTER EXPERT: Utilizes specialized skills, experience, and knowledge.
- COMMUNICATOR: Communicates effectively with the project manager, project team, and stakeholders.

**Assigned Duties**

- Achieves the project objective, working closely with the PM and project team.
- Applies Project Management and Agile concepts, methodology, and best practices.
- Works well in a variety of teams: Co-located, virtual, agile, and/or cross-functional teams.
- Delivers on commitments.

**Limits of Authority**

- Takes action and contributes to decisions, within the parameters of the project (cost, schedule, scope, quality).
- Explores and suggests alternatives for solving problems, keeping the project and team moving forward.
- Accountable to and reports to the Project Manager. Keeps Functional Manager informed.

*Source:* Connie Plowman, PMP, PMI Eric Jenett Project Management Excellence Award Recipient


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# 9.3a Identify Potential Resources (1 of 2)

- **Work functions**
  *(may include job titles and range of responsibilities)*
- **Professional discipline**
  *(may include degrees and professional certifications)*
- **Skill level**
  *(may include experience and performance ratings)*
- **Physical location**
  *(may include willingness to relocate and travel)*
- **Organizational/administrative unit**
  *(may include costs and contractual issues)*


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# 9.3a Identify Potential Resources (2 of 2)

**Resource Breakdown Structure** (RBS) is a grouping of all resources into main categories in level one and populating each main category with resources based on either function or skill level.


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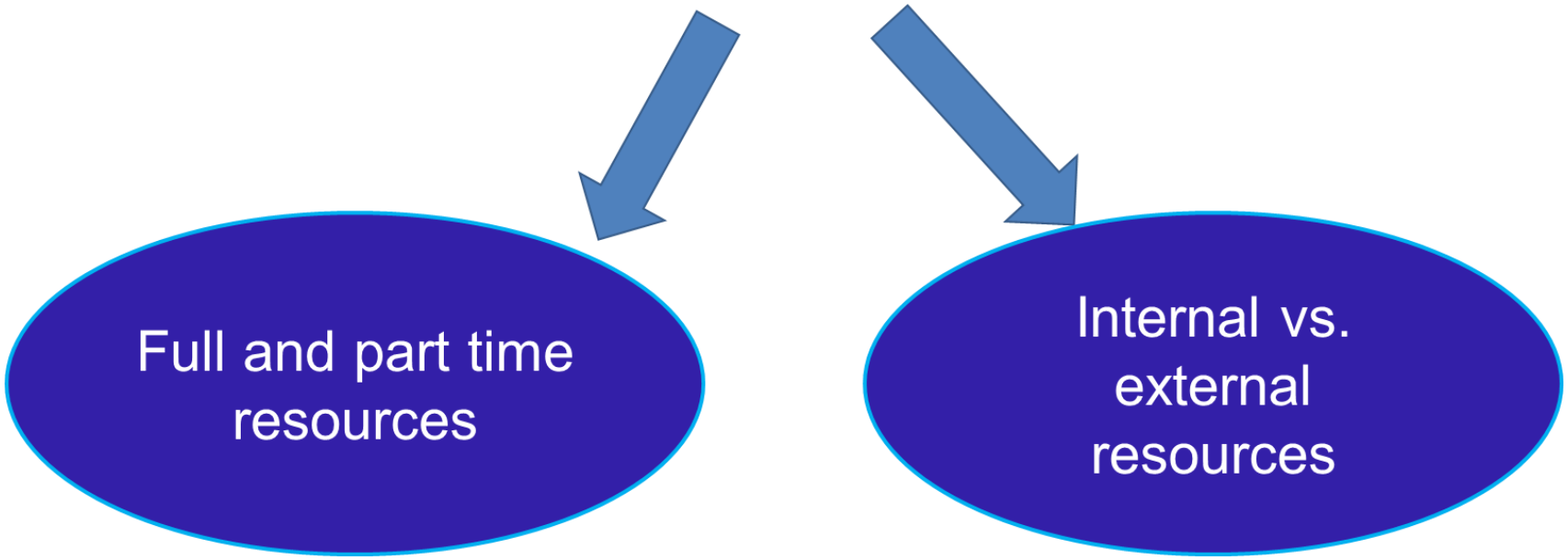
# Exhibit 9.4

## Resource Breakdown Structure Example

| | | | Unit | Rate |
|---|---|---|---|---|
| **1.0** | **Personnel** | | | |
| | 1.1 Management | | | |
| | 1.1.1 | Project Manager | hour | $100.00 |
| | 1.1.2 | Project Engineer | hour | $ 80.00 |
| | 1.2 Design | | | |
| | 1.2.1 | Civil Engineer | hour | $ 80.00 |
| | 1.2.2 | HVAC Engineer | hour | $ 80.00 |
| | 1.3 Construction | | | |
| | 1.3.1 | Foreman | hour | $ 70.00 |
| | 1.3.2 | Draftsman | hour | $ 50.00 |
| | 1.3.2 | Electrician | hour | $ 60.00 |
| | 1.3.3 | Carpenter | hour | $ 40.00 |
| | 1.3.4 | Mason | hour | $ 35.00 |
| **2.0** | **Materials** | | | |
| | 2.1 Civil/Architectural | | | |
| | 2.1.1 | Lumber | each | $ 4.00 |
| | 2.1.2 | Paint | gallon | $ 20.00 |
| | 2.1.3 | Drywall (0.5″ × 4′ × 8″) | each | $ 8.50 |
| | 2.2 Electrical | | | |
| | 2.2.1 | Wire | foot | $ 0.75 |
| | 2.2.2 | Switches | each | $ 4.75 |
| | 2.2.3 | Lighting fixtures | each | $ 55.00 |


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# 9.3b Determine Resource Availability





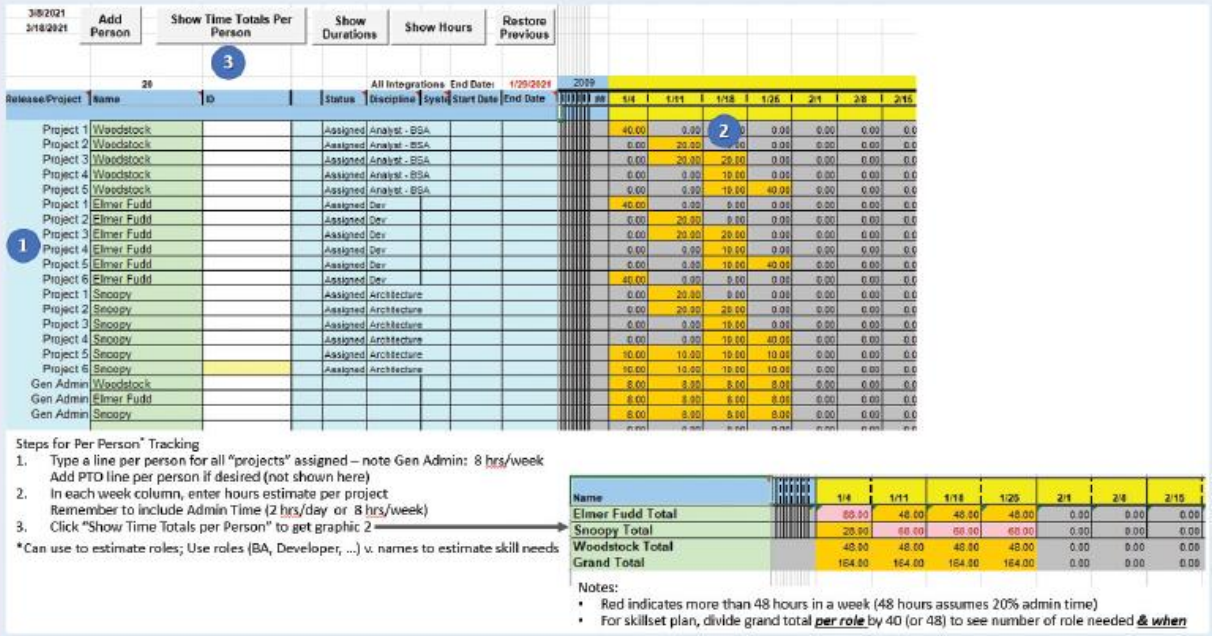
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# Exhibit 9.5

## Managing Resource Availability

This is an Excel spreadsheet used for project or department resource management. "Resource": Persons by name or roles (BA, Developer, Architect…) based on skill sets for a project or within a department. The graphic steps are the process. The spreadsheet has a "Resource Plan-Detail" tab and an "Allocation" tab. Enter data on the "Resource Plan-Detail" tab. Do step 3 to auto-generate the "Allocation" tab for ***actionable information***. Plan in detail as far out as needed or in a rolling fashion.

**Graphic:**




*Source:* John Stark


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# 9.4 Project Team Composition Issues

| |
|---|
| 9.4a Cross-Functional Teams |
| 9.4b Co-Located Teams |
| 9.4c Virtual Teams |
| 9.4d Outsourcing |


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# 9.4a Cross-Functional Teams

Projects typically require inputs from multiple disciplines and, therefore, require cross-functional teams. A Project Manager may need to:

- **become a mediator**
  *among various functions that are working on the project.*
- **understand and speak effectively with people**
  *representing different disciplines and various technical experts.*


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# 9.4b Co-Located Teams

- Co-location helps to create these opportunities for easy and quick communications, relationship building, teamwork, and productivity improvement.
- A major and important advantage of co-located teams is networking and an informal structure that **promote a work culture of sharing, collaborating, and supporting** each other.


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# 9.4c Virtual Teams

- Sometimes, a project requires the expertise of many people who are geographically dispersed, and it is impractical to have them all work in the same location or proximity.
- These teams require many forms of communication.


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# 9.4d Outsourcing

- Project Managers often are faced with the prospect of not finding the necessary talent within their organization.
- Outsourcing can allow a project to bring in talent from anywhere in the world.


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# 9.5 Assign a Resource to Each Activity

| |
|---|
| 9.5a Show Resource Responsibilities on RACI Chart |
| 9.5b Show Resource Assignments on Gantt Chart |
| 9.5c Summarize Resource Responsibilities by Time Period with Histogram |


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# 9.5a Show Resource Responsibilities on RACI Chart

**Responsibility Assignment Matrix** (RAM) is a matrix depicting work packages and assigned resources for completing activities required to produce work packages.

**RACI** is a popular form of RAM that presents roles of key Stakeholders and their roles defined as responsible (**R**), accountable (**A**), consult (**C**), and inform (**I**) for project activities in a matrix form.

*Agile team members decide among themselves who will do each work activity; when finished, they begin next-highest-priority work.*


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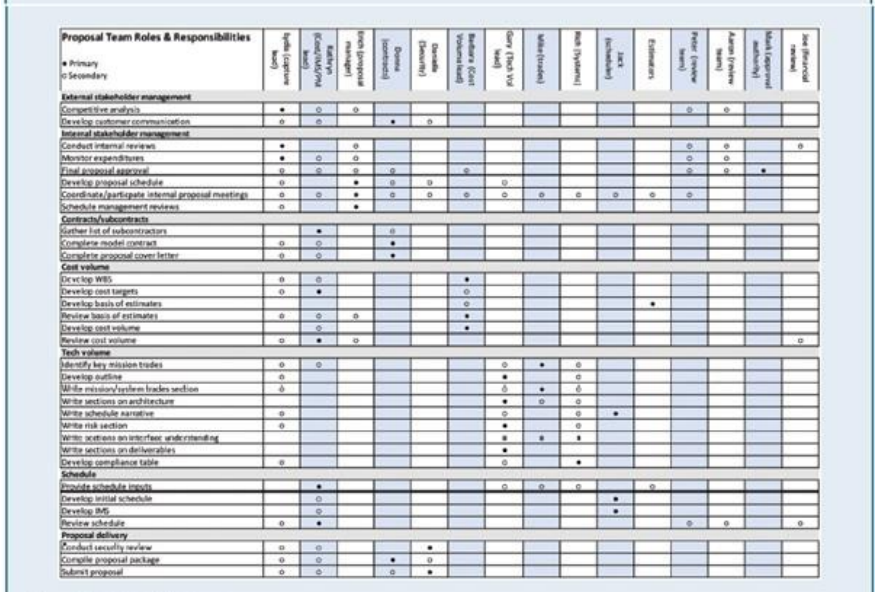
# Exhibit 9.6

## RACI Chart

The aerospace industry competes for work from the U.S. Government through competitive solicitations. As such, our work relies on a bid and proposal process to start our projects. The government releases a Request for Proposal (RFP) that contains details on the scope of their project and contract, and prospective bidders respond with proposals describing their technical solution as to how they plan to achieve the goals of the project and what that will cost.

Developing a proposal is in essence a project unto itself, with scope defined by the RFP, a budget to support the efforts associated with preparing the proposal, and a schedule determined by the proposal due date. Proposal development efforts are generally short, from two weeks to two months, so team members need to understand roles and responsibilities to effectively coordinate and communicate. As such, RACI charts are an effective communication tool for proposal projects. However, since proposal teams are comprised of people with broad backgrounds and varying degrees of project management background, we simplify the RACI construct to make it easier for everyone to understand.

For example, we use a form of the RACI chart as below for proposal projects, identifying primary and secondary roles for each task area. There can only be one primary person for each task, but there can be multiple secondary people. The primary person identified is the person who has the authority to approve and complete the task, which combines the "R" and "A" roles in a traditional RACI chart. We identify people that are involved in the task in any form as secondary, which would be the "C" and "I" roles. Combined with a proposal team organizational chart, this helps everyone understand their role and who are the approval authorities for interim products.

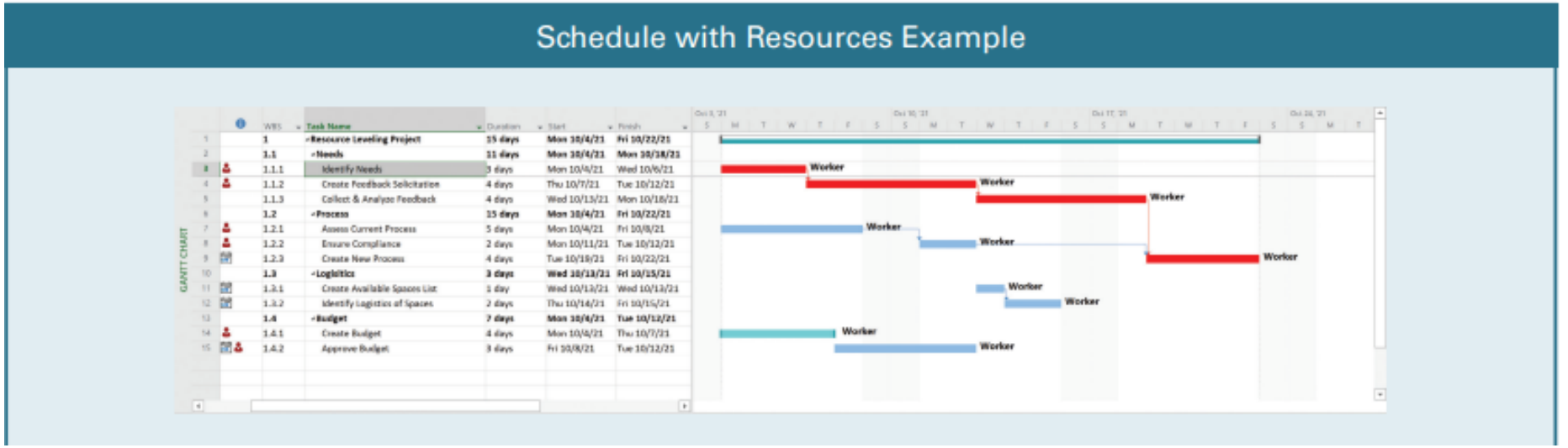


| Proposal Team Roles & Responsibilities<br>● Primary<br>○ Secondary | Ayala (capture lead) | Bill (Cost/VAM/Prod lead) | Kathryn (Contracts manager) | Erick (proposal manager) | Donna (contracts) | Danielle (Security) | Barbara (Cost volume lead) | Gary (Tech Vol lead) | Mike (trades) | Rob (Systems) | Jack (scheduler) | Estimators | Peter (review team) | Aaron (review team) | Mark (Approval authority) | Joe (financial review) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **External stakeholder management** | | | | | | | | | | | | | | | | |
| Competitive analysis | ● | ○ | ○ | | | | | | | | | | ○ | ○ | | |
| Develop customer communication | ○ | ○ | | ● | ○ | | | | | | | | | | | |
| **Internal stakeholder management** | | | | | | | | | | | | | | | | |
| Conduct internal reviews | ● | | ○ | | | | | | | | | | ○ | ○ | | ○ |
| Monitor expenditures | ● | ○ | ○ | | | | | | | | | | ○ | ○ | | |
| Final proposal approval | ○ | ○ | ○ | ○ | | | ○ | | | | | | ○ | ○ | ● | |
| Develop proposal schedule | ○ | | ● | ○ | ○ | | | ○ | | | | | | | | |
| Coordinate/participate internal proposal meetings | ○ | ○ | ● | ○ | ○ | | ○ | ○ | ○ | ○ | ○ | ○ | ○ | | | |
| Schedule management reviews | ○ | | ● | | | | | | | | | | | | | |
| **Contracts/subcontracts** | | | | | | | | | | | | | | | | |
| Gather list of subcontractors | | ● | | ○ | | | | | | | | | | | | |
| Complete model contract | ○ | ○ | | ● | | | | | | | | | | | | |
| Complete proposal cover letter | ○ | ○ | | ● | | | | | | | | | | | | |
| **Cost volume** | | | | | | | | | | | | | | | | |
| Develop WBS | ○ | ○ | | | | | ● | | | | | | | | | |
| Develop cost targets | ○ | ● | | | | | ○ | | | | | | | | | |
| Develop basis of estimates | | | | | | | ○ | | | | | ● | | | | |
| Review basis of estimates | ○ | ○ | ○ | | | | ● | | | | | | | | | |
| Develop cost volume | | ○ | | | | | ● | | | | | | | | | |
| Review cost volume | ○ | ● | ○ | | | | | | | | | | | | | ○ |
| **Tech volume** | | | | | | | | | | | | | | | | |
| Identify key mission trades | ○ | ○ | | | | | | ○ | ● | ○ | | | | | | |
| Develop outline | ○ | | | | | | | ● | | ○ | | | | | | |
| Write mission/system trades section | ○ | | | | | | | ○ | ● | ○ | | | | | | |
| Write sections on architecture | | | | | | | | ● | ○ | ○ | | | | | | |
| Write schedule narrative | ○ | | | | | | | ○ | | ○ | ● | | | | | |
| Write risk section | ○ | | | | | | | ● | | ○ | | | | | | |
| Write sections on interface understanding | | | | | | | | ■ | ■ | ■ | | | | | | |
| Write sections on deliverables | | | | | | | | ● | | | | | | | | |
| Develop compliance table | ○ | | | | | | | ○ | | ● | | | | | | |
| **Schedule** | | | | | | | | | | | | | | | | |
| Provide schedule inputs | | ● | | | | | | ○ | ○ | ○ | | ○ | | | | |
| Develop initial schedule | | ○ | | | | | | | | | ● | | | | | |
| Develop IMS | | ○ | | | | | | | | | ● | | | | | |
| Review schedule | ○ | ● | | | | | | | | | | | ○ | ○ | | ○ |
| **Proposal delivery** | | | | | | | | | | | | | | | | |
| Conduct security review | ○ | ○ | | | ● | | | | | | | | | | | |
| Compile proposal package | ○ | ○ | | ● | ○ | | | | | | | | | | | |
| Submit proposal | ○ | ○ | | ○ | ● | | | | | | | | | | | |

*Source:* Lydia Lavigne, Ball Aerospace.


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# 9.5b Show Resource Assignments on Gantt Chart

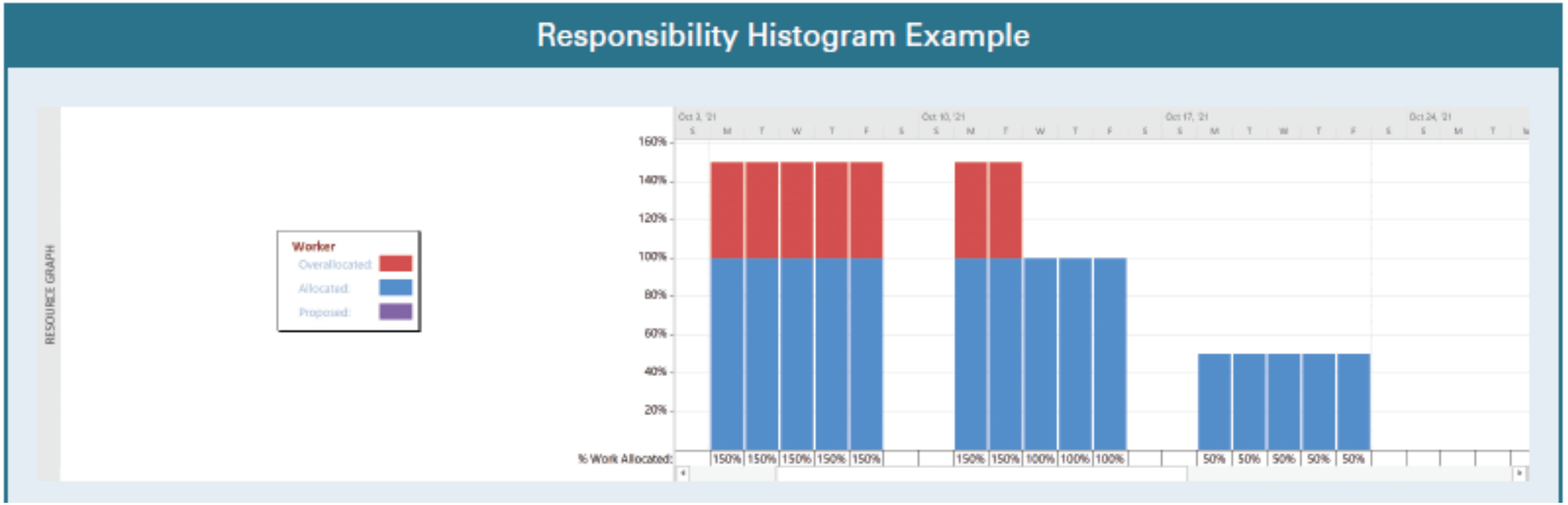



*Source:* Microsoft product screenshots reprinted with permission from Microsoft Corporation.

*The person responsible for each activity is listed next to the activity in the Gantt Chart Schedule.*


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# Exhibit 9.8




*Source:* Microsoft Corporation.

*Add up demands for each worker at each time period.*


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# 9.6 Dealing with Resource Overload

9.6a Methods of Resolving Resource Overload

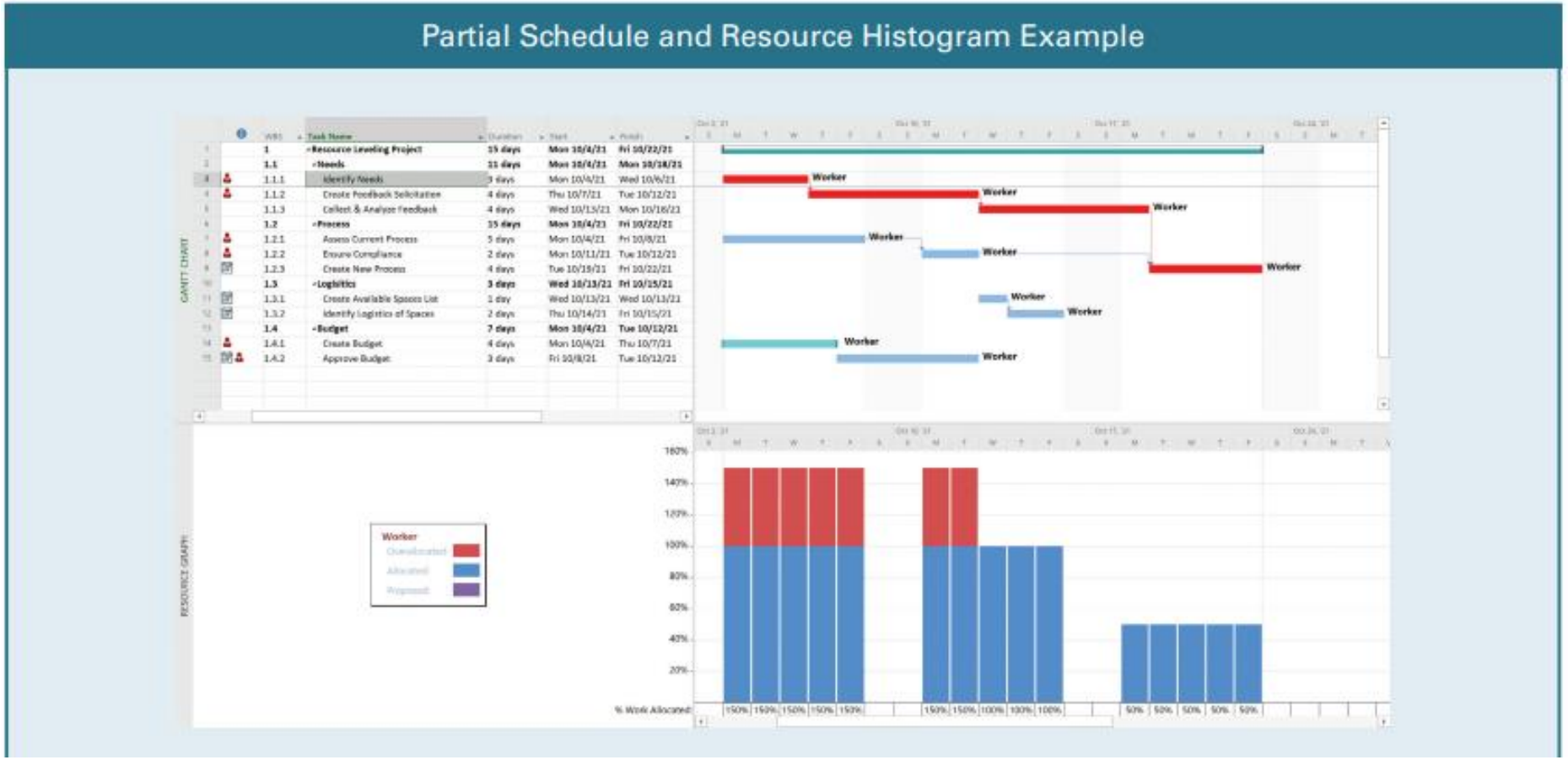

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# 9.6 Dealing with Resource Overloads

- Identify which activities are involved.
- Compare resource histogram to Gantt Chart Schedule.
- Project scheduling software pinpoints when overloads occur for each worker.
- Management decisions required to solve the problem.


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# Exhibit 9.9

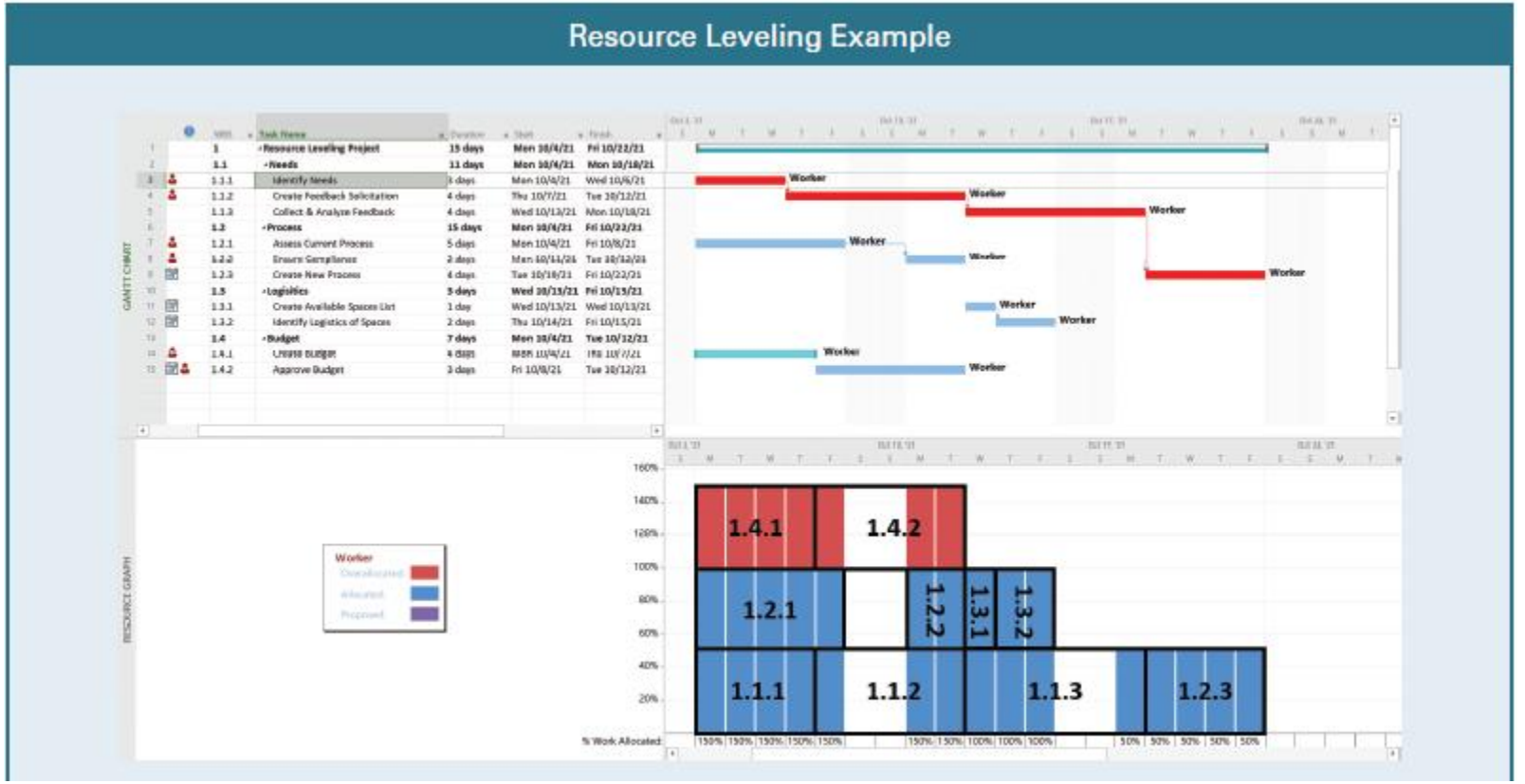

Partial Schedule and Resource Histogram Example
GANTT CHART
RESOURCE GRAPH
Worker
Worker
Worker

# 9.6a Methods of Resolving Resource Overloads

- Reorder activities.
- Acquire or borrow additional resources.
- Reduce project scope or extend project schedule.
- Inform sponsor of severe overloads.
- Resource-level the overloaded person's schedule.
- Assign certain activities to other workers.
- Split an activity into two activities:
  - *Perform first part as scheduled.*
  - *Delay second part of activity.*


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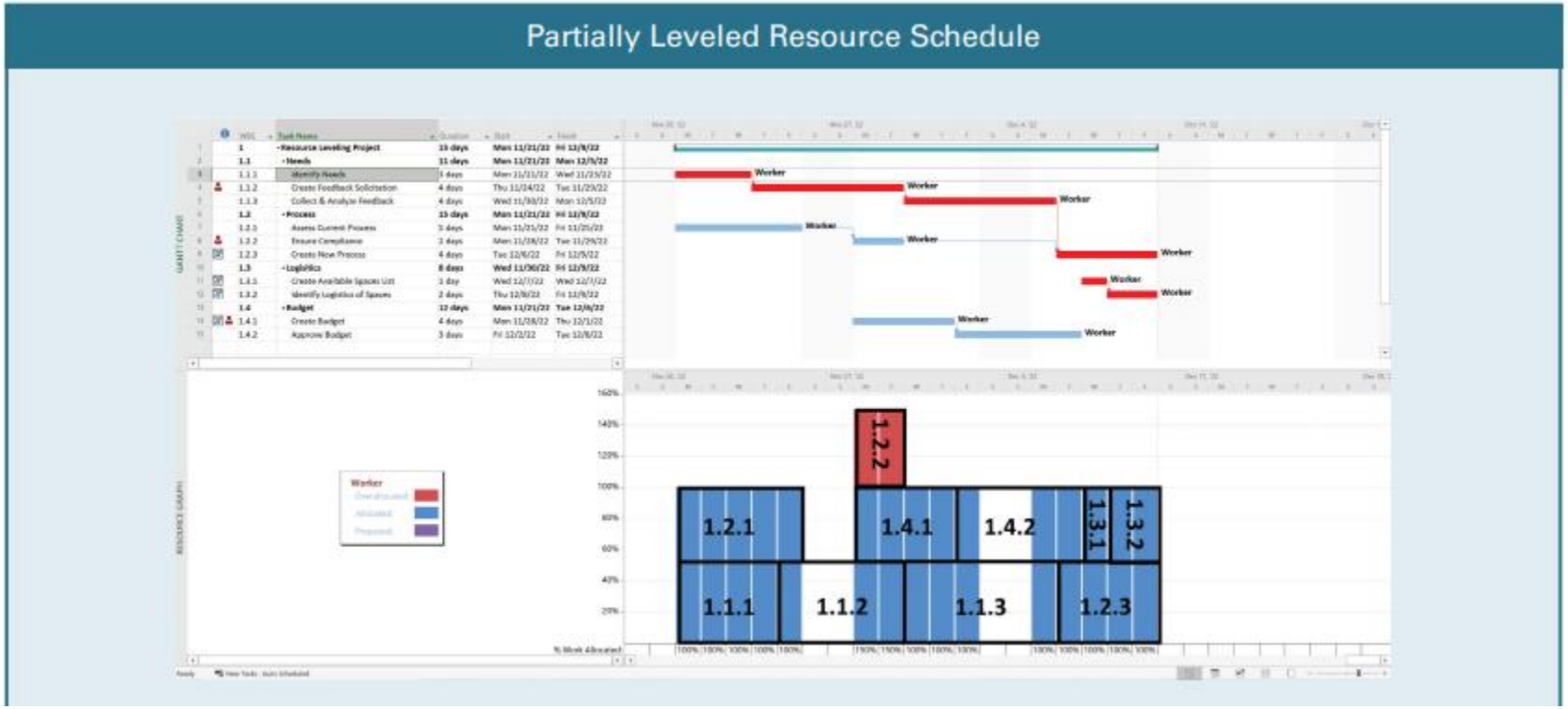

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# 9.6a Resource Leveling (1 of 2)

**Resource Leveling** is a project execution technique of adjusting the use of resources based on availability and the amount of float on activities to accomplish work as soon as possible, given the limited resource availability.


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# Exhibit 9.11




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# 9.6a Resource Leveling (2 of 2)

- Resource leveling often reduces an overload, but it does not always eliminate it entirely.
- If the non-critical activities must be completed at the rate of effort shown in the original schedule, some of them may need to be assigned to another worker.
- Resource leveling can be as much art as science.


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# 9.7 Compress the Project Schedule

| |
|---|
| 9.7a Actions to Reduce the Critical Path |
| 9.7b Crashing |
| 9.7c Fast Tracking |


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# 9.7a Actions to Reduce the Critical Path

- Reduce project scope and/or quality.
- Overlap sequential activities.
- Partially overlap sequential activities.
- Increase number of work hours/days.
- Schedule activities at same time.
- Shorten activities by assigning more resources.
- Shorten activities that cost least to speed up.


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# 9.7a Compress the Project Schedule

**Crashing** is speeding up the critical path often by adding additional resources or employing existing resources for longer hours and/or more days per week.

**Fast Tracking** is a method to expedite a project by executing activities at the same time that would ordinarily be done one after the other.

*In Agile, crashing is called "swarming," since team members swarm on a problem to help get back on track.*


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# 9.7b Crashing

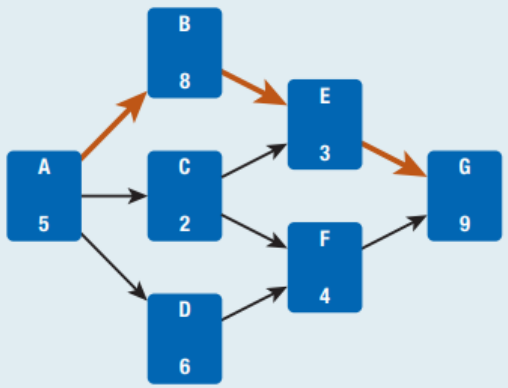
When deciding which activities to speed up, two questions must be asked:

1. Which activities are on the critical path?
2. Which critical path activity costs the least on a per-day basis to speed up?


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# Exhibit 9.12

## Crashing Example Set-Up




Legend

Activity Code

Duration

| Path | Duration |
|---|---|
| ABEG | 25 |
| ACEG | 19 |
| ACFG | 20 |
| ADFG | 24 |

| Activity | Normal Time | Normal Cost | Crash Time | Crash Cost | Crash Cost per Day |
|---|---|---|---|---|---|
| ▸ A | 5 | $300 | 5 | $300 | N/A |
| ▸ B | 8 | 250 | 5 | 400 | 50 |
| C | 2 | 100 | 2 | 100 | N/A |
| D | 6 | 300 | 3 | 600 | 100 |
| ▸ E | 3 | 150 | 2 | 300 | 150 |
| F | 4 | 275 | 3 | 300 | 25 |
| ▸ G | 9 | 700 | 8 | 900 | 200 |

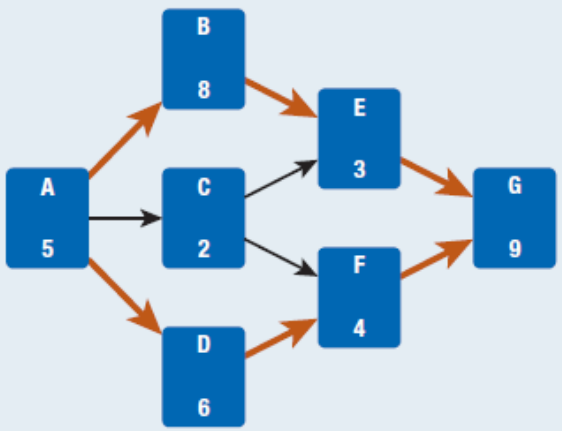
| Project Duration | Activity(IES) Crashed | Incremental Crash Cost | Cumulative Crash Cost |
|---|---|---|---|
| 25 | - | 0 | 0 |

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# Exhibit 9.13

## Crashing Example After One Round




Legend

Activity Code

Duration

| Path | Duration |
|---|---|
| ABEG | 25, 24 |
| ACEG | 19 |
| ACFG | 20 |
| ADFG | 24 |

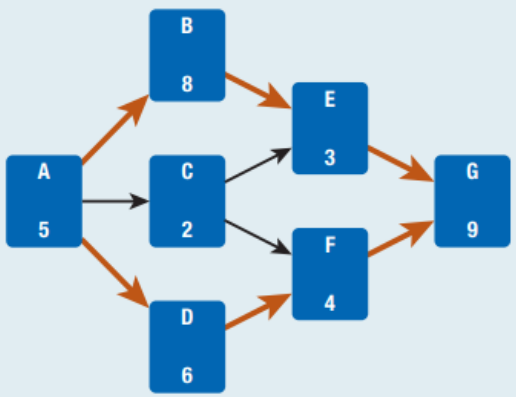
| Activity | Normal Time | Normal Cost | Crash Time | Crash Cost | Crash Cost per Day |
|---|---|---|---|---|---|
| ▸ • A | 5 | $300 | 5 | $300 | N/A |
| ▸ B | 8, 7 | 250 | 5 | 400 | 50 |
| C | 2 | 100 | 2 | 100 | N/A |
| • D | 6 | 300 | 3 | 600 | 100 |
| ▸ E | 3 | 150 | 2 | 300 | 150 |
| • F | 4 | 275 | 3 | 300 | 25 |
| ▸ • G | 9 | 700 | 8 | 900 | 200 |

| Project Duration | Activity(IES) Crashed | Incremental Crash Cost | Cumulative Crash Cost |
|---|---|---|---|
| 25 | - | 0 | 0 |
| 24 | B | $50 | $50 |


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# Exhibit 9.14

Crashing Example After Two Rounds




Legend

Activity Code

Duration

| Path | Duration |
|---|---|
| ABEG | ~~25~~ ~~24~~ 23 |
| ACEG | 19 |
| ACFG | ~~20~~ 19 |
| ADFG | ~~24~~ 23 |

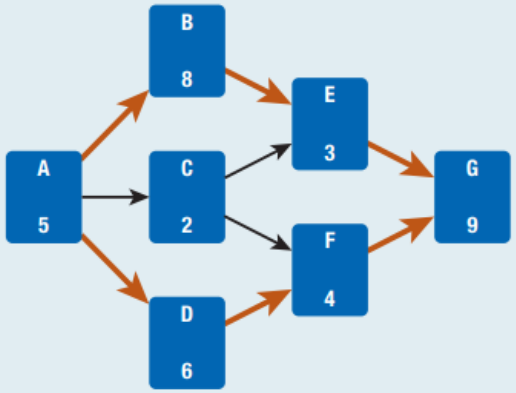
| Activity | Normal Time | Normal Cost | Crash Time | Crash Cost | Crash Cost per Day |
|---|---|---|---|---|---|
| ▸ • ~~A~~ | ~~5~~ | ~~$300~~ | ~~5~~ | ~~$300~~ | ~~N/A~~ |
| ▸ B | ~~8~~ ~~7~~ 6 | 250 | 5 | 400 | 50 |
| ~~C~~ | ~~2~~ | ~~100~~ | ~~2~~ | ~~100~~ | ~~N/A~~ |
| • D | 6 | 300 | 3 | 600 | 100 |
| ▸ E | 3 | 150 | 2 | 300 | 150 |
| • ~~F~~ | ~~4~~ 3 | ~~275~~ | ~~3~~ | ~~300~~ | ~~25~~ |
| ▸ • G | 9 | 700 | 8 | 900 | 200 |

| Project Duration | Activity(IES) Crashed | Incremental Crash Cost | Cumulative Crash Cost |
|---|---|---|---|
| 25 | - | 0 | 0 |
| 24 | B | $50 | $50 |
| 23 | B & F | 75 (50+25) | 125 |


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# Exhibit 9.15

## Crashing Example in All-Crash Mode




Legend

Activity Code

Duration

| Path | Duration |
|---|---|
| ABEG | ~~25~~ ~~24~~ ~~23~~ ~~22~~ ~~21~~ 20 |
| ACEG | ~~19~~ ~~18~~ 17 |
| ACFG | ~~20~~ ~~19~~ 18 |
| ADFG | ~~24~~ ~~23~~ ~~22~~ ~~21~~ 20 |

| Activity | Normal Time | Normal Cost | Crash Time | Crash Cost | Crash Cost per Day |
|---|---|---|---|---|---|
| • A | 5 | $300 | 5 | $300 | N/A |
| B | ~~8~~ ~~7~~ ~~6~~ 5 | 250 | 5 | 400 | 50 |
| C | 2 | 100 | 2 | 100 | N/A |
| • D | ~~6~~ ~~5~~ 4 | 300 | 3 | 600 | 100 |
| E | ~~3~~ 2 | 150 | 2 | 300 | 150 |
| • F | ~~4~~ 3 | 275 | 3 | 300 | 25 |
| • G | ~~9~~ 8 | 700 | 8 | 900 | 200 |

| Project Duration | Activity(IES) Crashed | Incremental Crash Cost | Cumulative Crash Cost |
|---|---|---|---|
| 25 | - | 0 | 0 |
| 24 | B | $50 | $50 |
| 23 | B & F | 75 (50+25) | 125 |
| 22 | B & D | 150 (50+100) | 275 |
| 21 | G | 200 | 475 |
| 20 | D & E | 250 (100+150) | 725 |

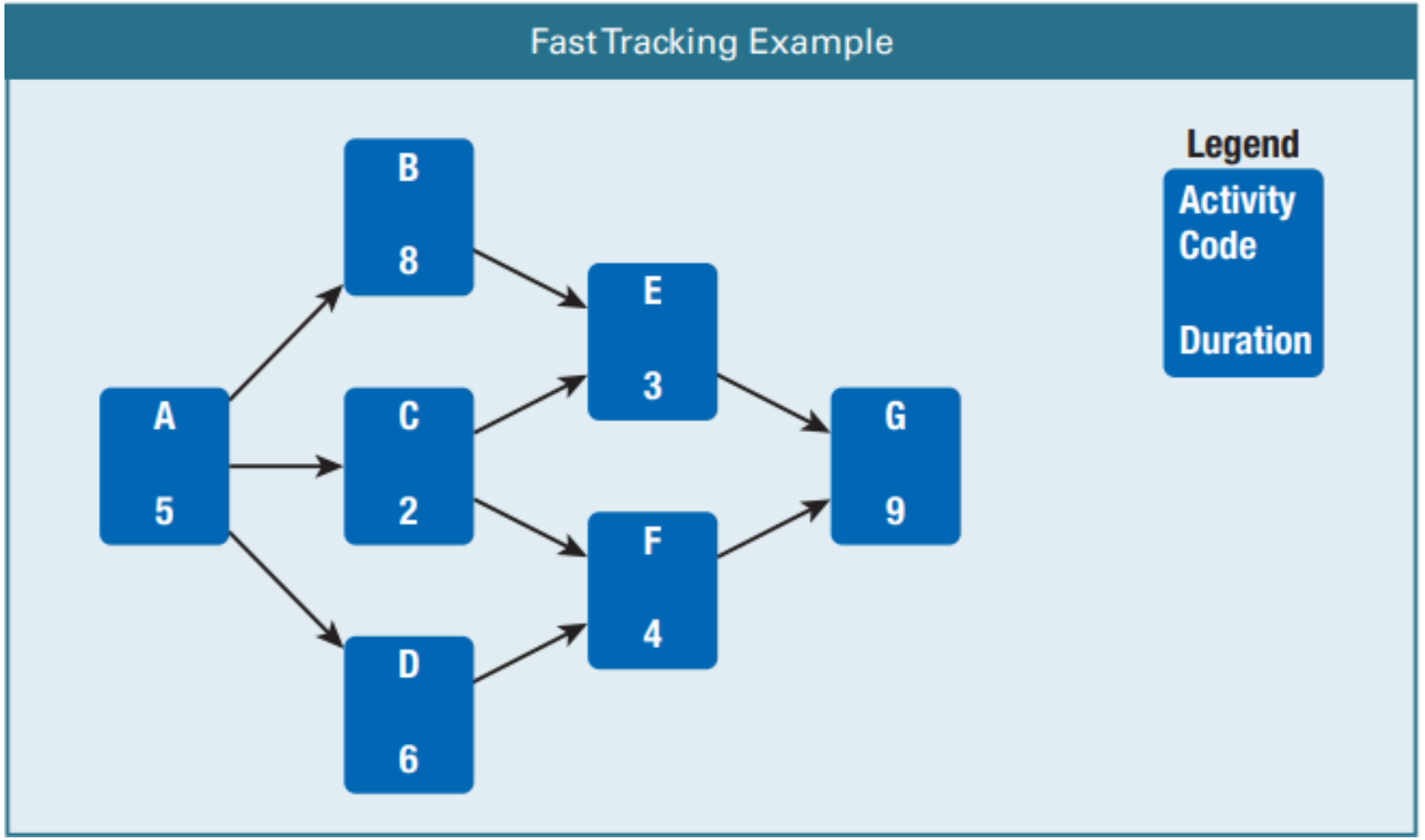

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# 9.7c Fast Tracking

Fast tracking occurs when activities that are normally performed in series (one after the other) are performed at the same time.


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# Exhibit 9.16





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# 9.8 Alternative Scheduling Methods

| |
|---|
| 9.8a Critical Chain Project Management (CCPM) |
| 9.8b Reverse Phase Schedules |
| 9.8c Rolling Wave Planning |
| 9.8d Auto/Manual Scheduling |


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# 9.8a Critical Chain Project Management (CCPM)

**Critical Chain Method** is an alternative scheduling technique that modifies project schedule by taking resource constraints into account . . . allows the project team to place buffers on any project schedule path to account for uncertainty and limited resources.


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# 9.8b Reverse Phase Schedules

- A.k.a. Last Planner System.
- Developed by people closest to the work.
- Start with final project deliverables.
- Ask what needs to be completed prior to starting work on this deliverable.
- Think from end of the project to the beginning.
- Often used in construction projects.


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# 9.8c Rolling Wave Planning

- Plan first part of project in detail.
- Plan later parts at a high level/use placeholders.
- Focus on the near term.
- Extreme of rolling wave planning is Agile.


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# 9.8d Auto/Manual Scheduling

- MS Project now allows manual scheduling.
- Enables users to emulate MS Excel.
- When showing few milestones without dates *(i.e., during Chartering)*, manual scheduling may be good option.


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# 9.9 *PMBOK Guide 7e*

| |
|---|
| 9.9a Teams |
| 9.9b Development Approach and Life Cycle |
| 9.9c Project Work |


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# 9.9a Team

- The desired outcomes of the Team domain are a high-performing team, shared ownership, and “applicable leadership and interpersonal skills demonstrated by all team members.”
- Clarity is needed in terms of who has responsibility, accountability, and authority for various types of decisions.


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# 9.9b Development Approach and Life Cycle

- This approach will guide the resourcing of the project.
- Team members should be recognized for their contribution when their work is finished.
- A good Project Manager will spend time getting to know the motivations of their team members and help inspire them.


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# 9.9c Project Work

This approach will largely determine when and how work is executed and delivered which, in turn, dictates human resource requirements


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# 9.10 Agile Project Resourcing

| |
|---|
| Predictive versus Agile Project Resourcing |
| 9.10a Agile Terms |


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# 9.10 Predictive versus Agile Project Resourcing

- People are resourced differently in plan-driven and Agile projects.
- We reduce uncertainty in traditional project management *(using a plan-driven approach)* with a primary focus on following the plan.
- Agile projects are change-driven, and uncertainty is embraced.


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# Exhibit 9.17

## Predictive versus Agile Project Resourcing

| Project Resourcing and Accelerating Questions | Predictive | Agile |
|---|---|---|
| Ideally how long is the project team together? | Duration of project | Long-term |
| What is the more likely limiting schedule factor? | Activities and their logical order and estimated durations | Size and ability of the team |
| What is needed to get the team going? | Identify potential resources, decide timing, select, and on-board members | Keep the existing team going |
| How is work divided among team members? | Project manager assigns then finds and resolves overloads | Team decides who does what and supports each other as needed |
| How is a predictable amount of project work managed? | A worker is assigned to each activity, then overloads are identified and resolved | Teams determine their velocity, then manage their cadence and cycle time |
| How are schedule compression issues handled? | Reduce scope or critical path, often by fast tracking or crashing | Reduce scope by delivering MVP early and letting the client decide |


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# 9.10a Agile Terms Used in Resourcing and Accelerating Projects

| TERM | DEFINITION |
|---|---|
| **Velocity** | *The forecasted number of story points a team can do in one iteration.* |
| **Cadence** | *The rhythm of planning, performing, and evaluating each iteration.* |
| **Cycle Time** | *The time from when work starts on an item until it is delivered.* |
| **Swarming** | *Multiple team members work quickly together to solve a problem.* |
| **Release Decision** | *Determination that product and support are ready for transition and customer is ready to successfully use the product.* |


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# 9.11 Using MS Project for Resource Allocation

| |
|---|
| 9.11a Step 1: Defining Resources |
| 9.11b Step 2: Set Up a Resource Calendar |
| 9.11c Step 3: Assigning Resources |
| 9.11d Step 4: Finding Overallocated Resources |
| 9.11e Step 5: Dealing with Overallocations |
| 9.11f Step 6: Crashing a Critical Path Activity |


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# 9.11a Step 1: Defining Resources (1 of 4)

- Describe available resources in the MS Project database.
- As a single unit or a pool of like units.
- Resources include:
  - *People*
  - *Buildings*
  - *Materials*
  - *Facilities*
  - *Supplies*


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# 9.11a Step 1. Defining Resources (2 of 4)

- Enter resource information before time of assignment to a task.
- Minimum field requirements for defining resources:
  - *Name*
  - *Max Units*
- If costs are to be modeled, must provide:
  - *Rates*
  - *Cost per Use*
  - *Accrue At values*


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# 9.11a Step 1: Defining Resources (3 of 4)

## Max Units Field

- Determined before resource is assigned to activities.
- Defines availability of a resource for project work.
- Default 100%, but individual resources not usually 100% available.
- Considerations:
  - *What is considered project work?*
  - *What is not considered project work?*
  - *Organizational holidays accounted for in the Standard (Project) calendar.*
  - *Vacation days accounted for in each resource calendar.*

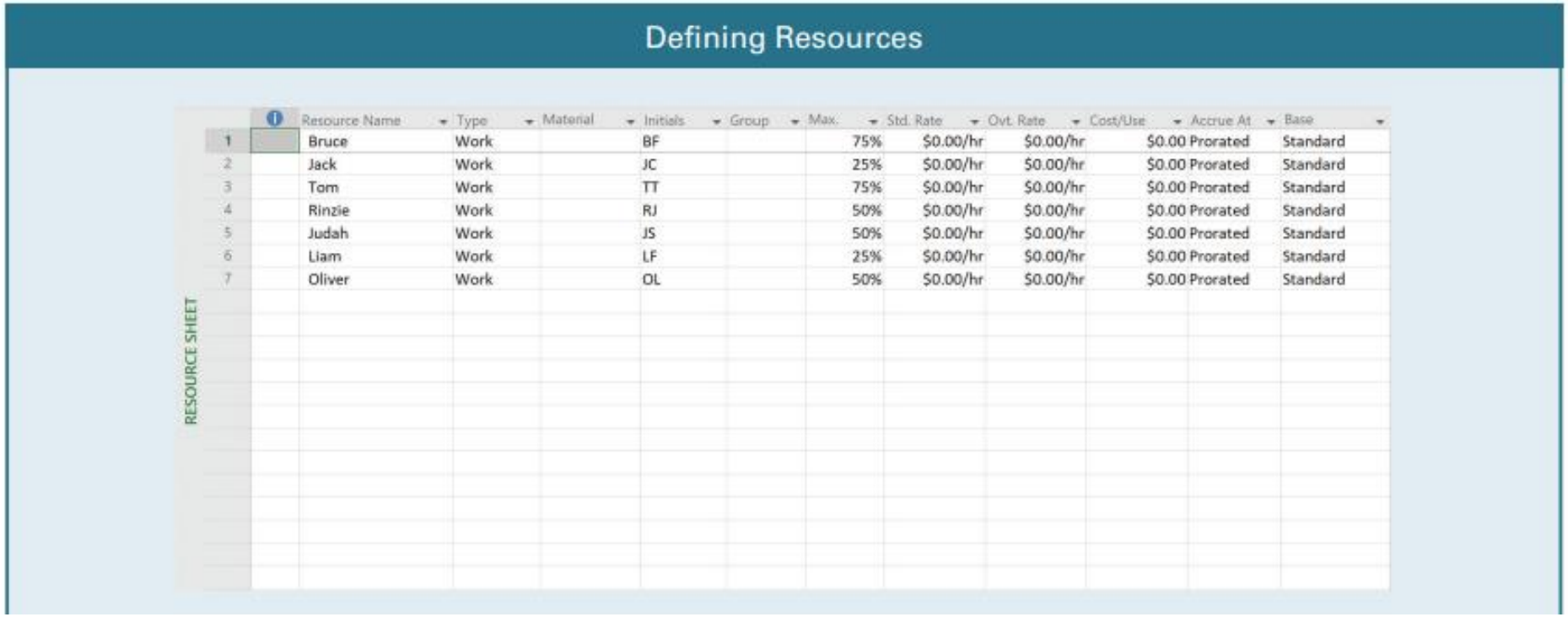

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# 9.11a Step 1: Defining Resources (4 of 4)

1. On View tab>> Resource Views group, click Resource Sheet.
2. In first blank row, enter resource name in Resource Name cell.
3. In Initials cell, enter initials of the resource.
4. Click Max Units cell and enter resource's maximum availability.


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# Exhibit 9.18




| | Resource Name | Type | Material | Initials | Group | Max. | Std. Rate | Ovt. Rate | Cost/Use | Accrue At | Base |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Bruce | Work | | BF | | 75% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 2 | Jack | Work | | JC | | 25% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 3 | Tom | Work | | TT | | 75% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 4 | Rinzie | Work | | RJ | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 5 | Judah | Work | | JS | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 6 | Liam | Work | | LF | | 25% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 7 | Oliver | Work | | OL | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |

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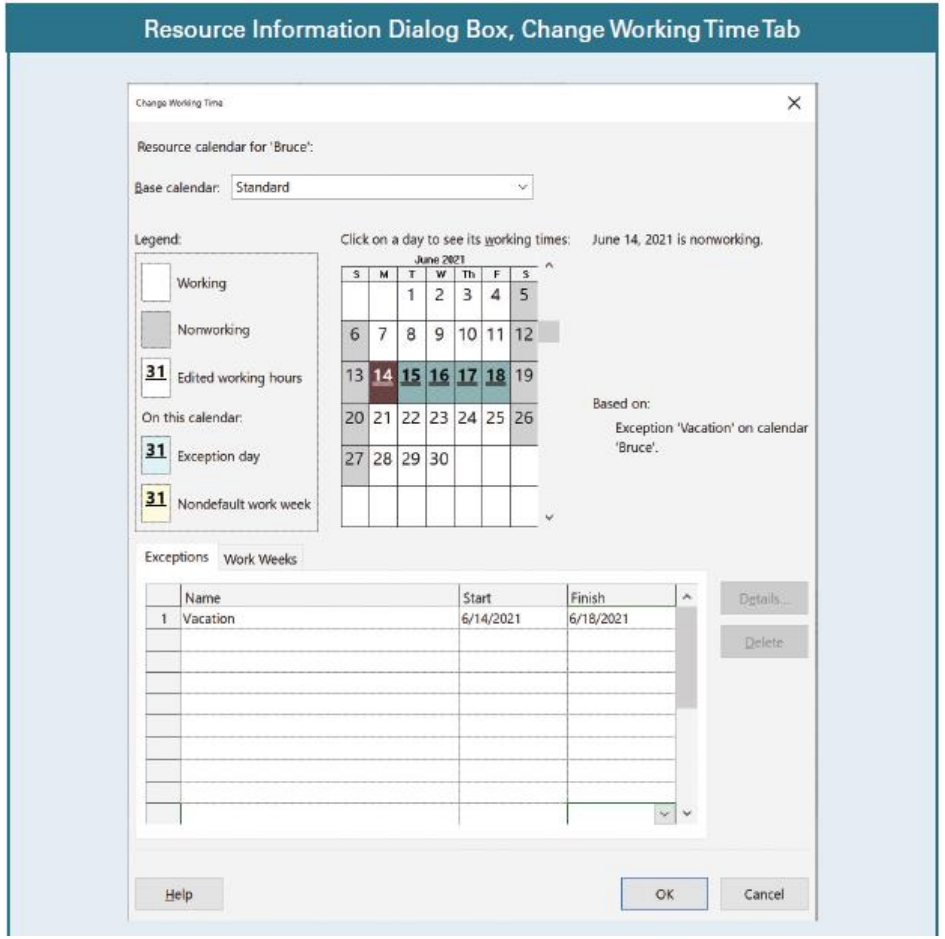

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# 9.11b Step 2: Set Up a Resource Calendar

1. Use to block out vacation days and other *resource-specific* non-working days.
2. **View Tab>>Resource Sheet.**
3. Double-click row of resource to activate dialog.
4. Click **Change Working Time …**
5. Make Revisions.


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# Exhibit 9.19




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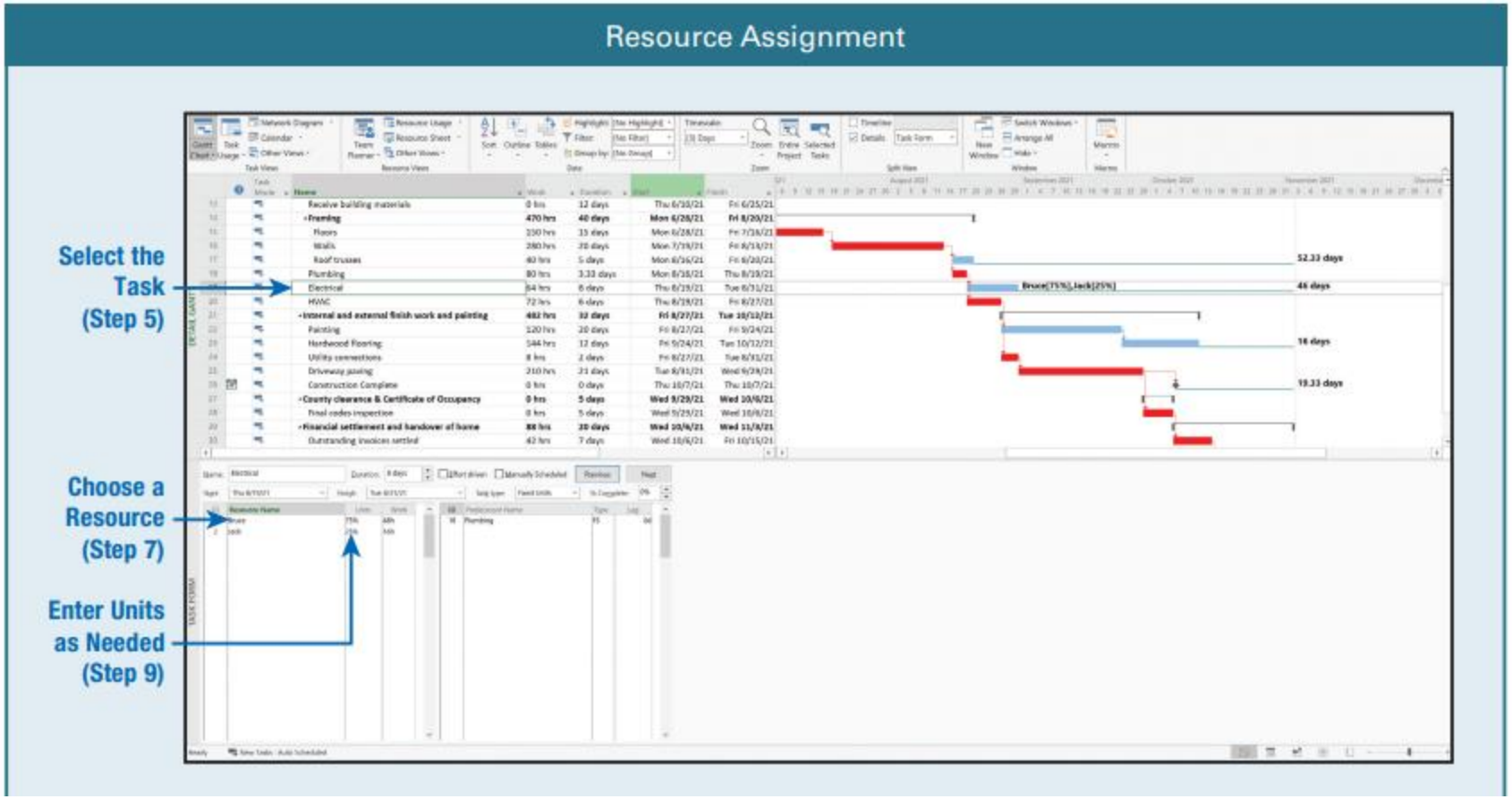
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# 9.11c Step 3: Assigning Resources

1. Click the **Task Tab**>>**View Group**>>**Gantt Chart**.
2. Right-click in the **Start** column header >>**Insert Column**>> type **Work**>> press **Enter** to add the Work column.
3. Click the **View Tab**>>**Split View Group**>> click **Details**>> choose **Task Form** in the drop-down list.
4. Right-click the form in the lower pane and select **Resources and Predecessors**.
5. In the upper pane, click the task row needing a resource assignment.
6. Click the first blank row in the **Resource Name** column in the lower pane's form.
7. Choose the resource name from the drop-down list.
8. Repeat Steps 6 and 7 to add additional resources to the assignment list.
9. Enter a **Units** value if the Max Units value is not correct for any assignment: *(If the Max Units entered in the Resource Sheet is correct, you can leave this blank, and it will autofill)*
10. Click the **OK** button *(no assignment is made until the OK button is clicked).*
11. Note that Work is calculated, and the activity duration value did not change.


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# Exhibit 9.20




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# 9.11c Step 3: Assigning Resources

- **Duration** *is the number of time units between the activity start and end (the default display value is in days spanning eight work hours).*
- **Units** *represents the availability of a resource for work each day.*
- **Work** *(hours assignment) is calculated by multiplying the Duration value (converted to hours) by the Units value.*
- **Task type** *determines which of three values (duration, units, and work) changes when one of the other two is modified (choices are Fixed Units, Fixed Duration, and Fixed Work).*


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# 9.11c Basic Assignment Calculation with Fixed Units Selected (Default)

- Activity with no prior resource assignments—Uses Duration and Units values and sums into Work field.
- Activity with one or more resources already assigned—MS Project holds Duration value constant and adjusts activity Work value.
- Removal of resources works in reverse.


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# 9.11c Modifying an Assignment

- MS Project maintains relationship among the Duration, Units, and Work values.
- If you don’t like MS Project adjusting Duration automatically based on resources, switch “Task type” to “Fixed Work.”

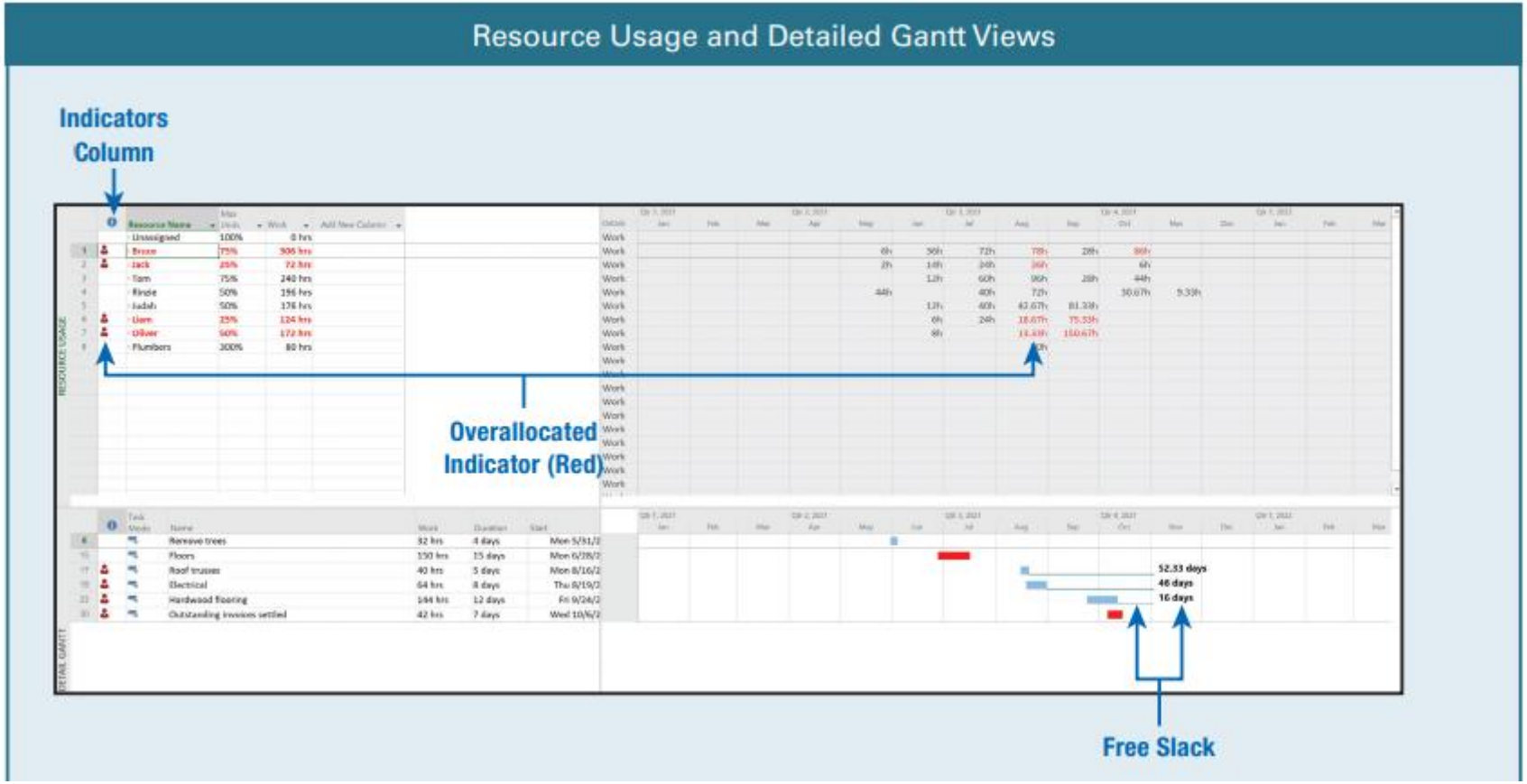

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# 9.11d Step 4: Finding Overallocated Resources (1 of 2)

1. Click the **Task Tab**>>**View Group**>>**Gantt Chart** drop-down menu >>click **More Views**…
2. On the **More Views dialog**>> scroll to **Resource Allocation**>> click **Edit**.
3. On the **View Definition dialog**>> click **Details Pane** drop-down >> choose **Detail Gantt**.
4. Click the **Show in menu** checkbox.
5. Click **OK**>> click **Apply**.
6. In the upper pane, right-click the **Work column header**>>**Insert Column**.
7. **Enter Max Units**>> click **Enter**.
8. In the lower pane, right-click the **Leveling Delay column header**>>**Insert Column**.
9. Enter **Work**>> click **Enter**.
10. In the upper pane, click the empty box to the left of the Indicators icon to select all rows.
11. Click the **View Tab**>>**Data Group**>>**Outline**>> **click Hide Subtasks**


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# Exhibit 9.21




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# 9.11d Step 4: Finding Overallocated Resources (2 of 2)

**To analyze overallocated resources in Resource Allocation View:**

1. Set timescale to the start of the schedule.
2. Slowly scroll timescale toward the end of the schedule.
3. Analyze each instance of cell values displayed in red for cause and severity.


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# 9.11e Step 5: Dealing with Overallocations

- Replace an overallocated resource with one that has time for the assignment.
- Reduce the Units assignment, extending the activity duration (this could affect the finish date of the project).
- Lessen the scope of one or more activities.
- Ignore the overload if the resource impact is temporary.
- Try **Resource Tab**>>**Level Group**>>**Level Resource** or **Level All** and see what happens to your timeline (remember, the Undo command is **Ctrl-Z**).


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# 9.11f Step 6: Crashing a Critical Path Activity (1 of 2)

1. Click the **View Tab**>>**Split View**>> uncheck **Details**.
2. Click the **View Tab**>>**Resource Sheet**.
3. In the next available row, type "**Plumbers**" in the **Resource Name** column.
4. In the **Initials** cell, enter the initials of the resource *(if different from the auto-generated).*
5. Click the **Max Units** cell and enter the resource's maximum availability as 300 percent. *(This means the project has three plumbers who can work 100 percent of the time on the project.)*
6. Your Resource Sheet should now look like Exhibit 9.22.
7. Click **the View Tab**>>**Task Views**>>**Gantt Chart**. *(Readjust timescale and column views as needed; ensure you can see the Start and Finish columns.)*
8. Double-click the **Plumbing** task>> **Resources Tab.**
9. In the first row of the Resource Name column, drop-down menu and choose **Plumbers**>> click **OK**.

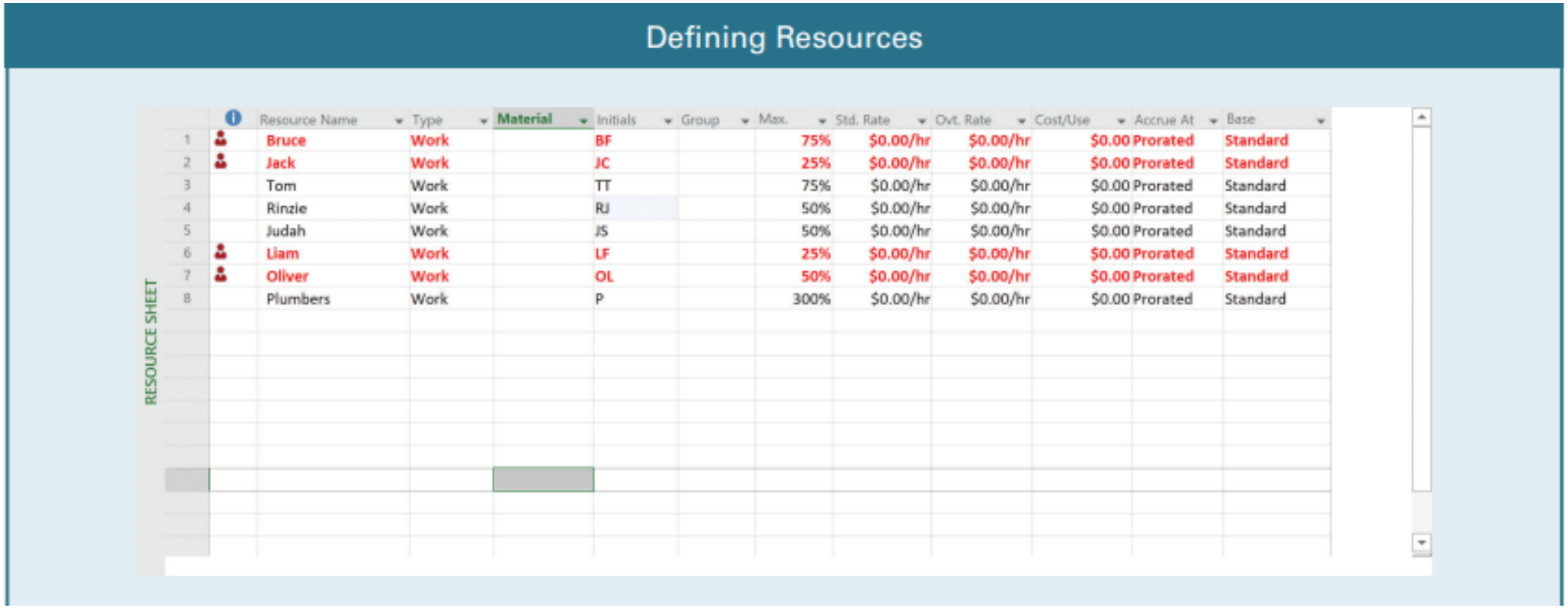

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# 9.11f Step 6: Crashing a Critical Path Activity (2 of 2)

10. This assigns one plumber to work on this task (100%).
11. You will notice work hours are added, but neither the critical path nor project's duration has changed.
12. Now assume that the Project Manager decides the Plumbing task needs to be completed more quickly, and the decision to crash that task is made.
13. Double-click on the **Plumbing** task >>**Resources Tab** again.
14. Change the Units from 100% to 300%. *(This will assign three plumbers to the task.)*
15. Click **OK**. *(See Exhibit 9.21.)*
16. You will now see that the critical path Gantt bar has shortened, the duration of the task has shortened, and the overall project finish date has shifted back (because Plumbing was a critical path activity).
17. Good job! You just got the plumbing done sooner!


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# Exhibit 9.22

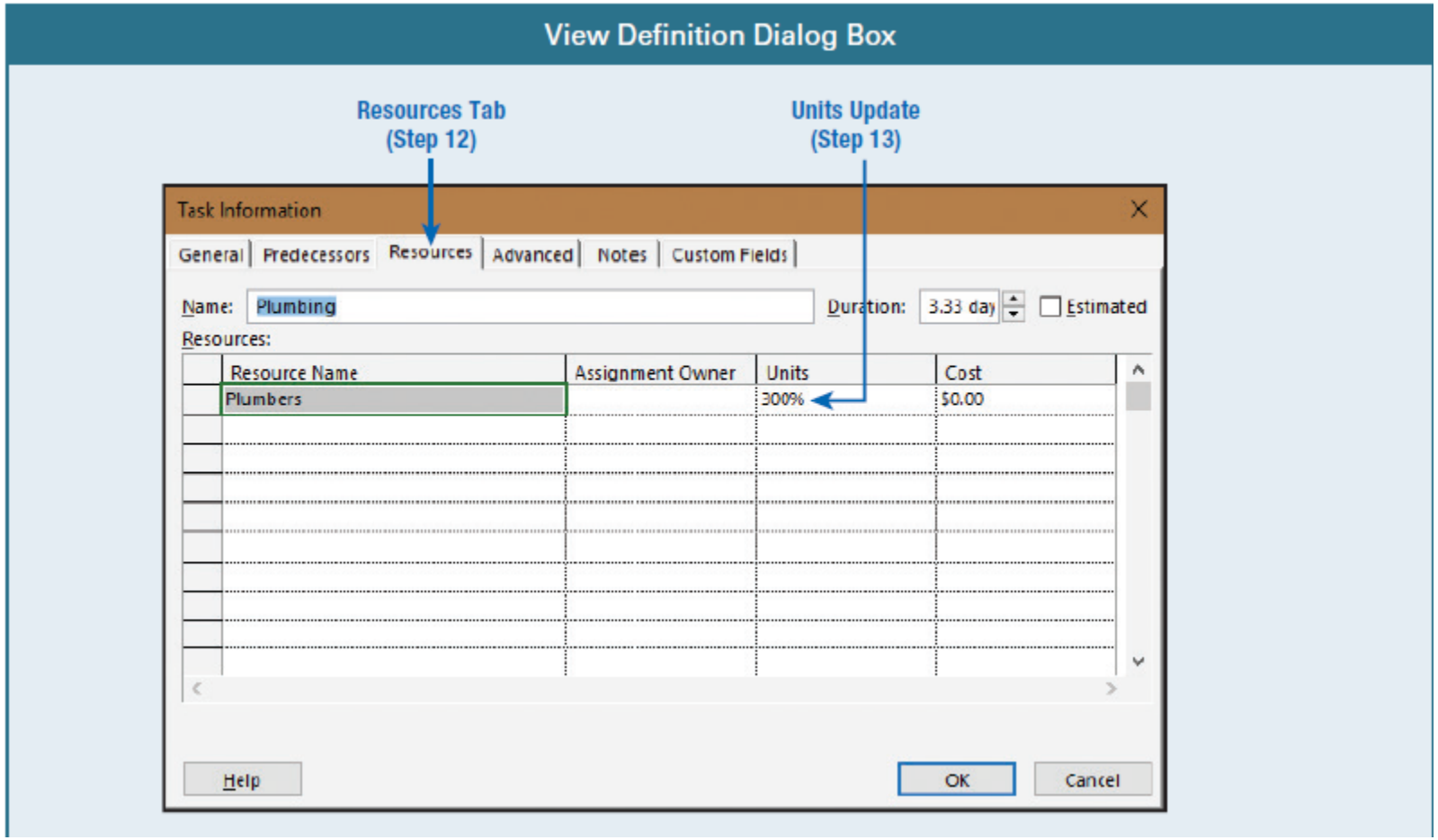



| | Resource Name | Type | Material | Initials | Group | Max. | Std. Rate | Ovt. Rate | Cost/Use | Accrue At | Base |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Bruce | Work | | BF | | 75% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 2 | Jack | Work | | JC | | 25% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 3 | Tom | Work | | TT | | 75% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 4 | Rinzie | Work | | RJ | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 5 | Judah | Work | | JS | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 6 | Liam | Work | | LF | | 25% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 7 | Oliver | Work | | OL | | 50% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |
| 8 | Plumbers | Work | | P | | 300% | $0.00/hr | $0.00/hr | $0.00 | Prorated | Standard |

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# Exhibit 9.23




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# Summary (1 of 2)

- PM needs to look at listed activities and estimate resources needed to perform each.
- Resource assignments often posted directly on a Gantt Chart schedule.
- RACI charts depict work activities on a vertical scale.
- Use responsibility histograms to determine overloads.
- Use critical path schedule, resource assignments, and the resource histogram to determine who is overloaded, when, and by what activities.
- Use resource leveling to reduce peak demands by postponing non-critical activities.


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# Summary (2 of 2)

- Look for methods of accelerating *(compressing)* the project schedule.
  - **Crashing**—decision to pay extra money to speed up certain activities on critical path
  - **Fast tracking**—allows activities normally conducted in sequence to be performed in parallel
- Alternative scheduling methods include: Critical chain, Reverse phase, Rolling wave, Agile, Auto/manual.
- Project scheduling software such as Microsoft Project 2016 is useful when determining resources for a project.


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